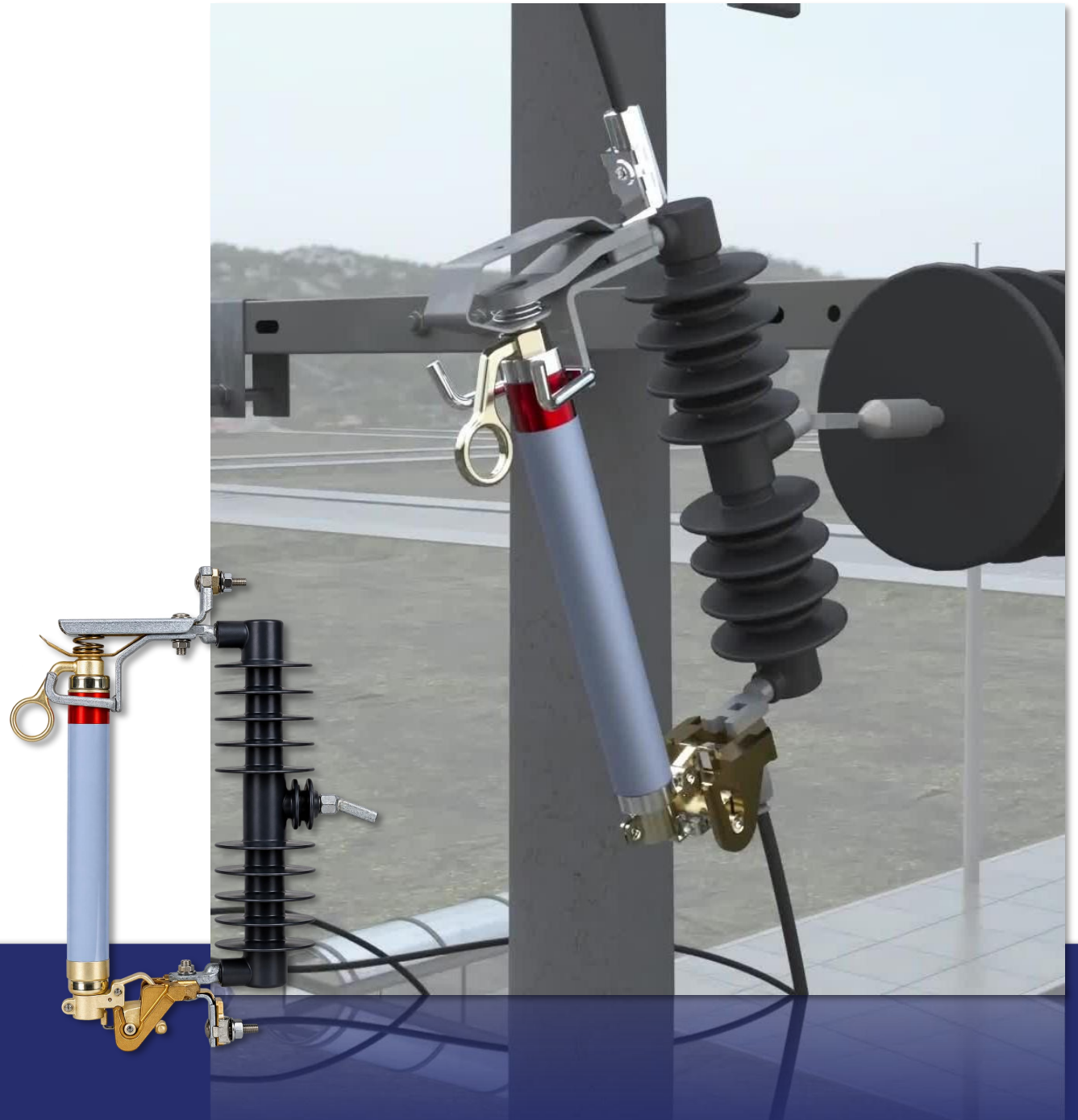
DUM 2026

REFORM TECH. CO.,Ltd.

Current-Limiting Drop-Out Fuse

Clearing faults without any external arcs



The distribution transformer challenge

Why this matters for Indian DISCOMs



Source: Central Electricity Authority (CEA), Standardisation Cell meeting, 29 Oct 2025

~10%

National distribution transformer failure rate

~1.3M

Transformer failures every year

>20%

Failure rate in some northern states

The fuse is the transformer's first line of protection.

A fuse can't fix how a transformer is built, but it decides how much fault energy reaches it.



High-speed footage · 25.8 kV / 12.5 kA breaking test

THE PROBLEM

Conventional fuses fail to clear the fault

Arc blown outside

Arc and hot gas are expelled from the tube

Open at the bottom

Moisture and salt degrade performance

Fitted by hand

Field assembly leaves room for error

One failed fuse, three problems

What follows when the fault isn't cleared

Wider outage

Fault current flows until an upstream recloser trips, so a whole area loses power.



Equipment damage

Longer faults mean more thermal and mechanical stress on transformers.



Fire risk

An exposed arc near dry grass, crops, or forest can start a fire.



Photo: Donghae Fire Station, 2019

THE SOLUTION

Reformtech's CLF Fuse

One sealed unit

Fuse holder and fuse link combined

Works like a standard drop-out fuse

Mounts on a cutout base, drops out visibly

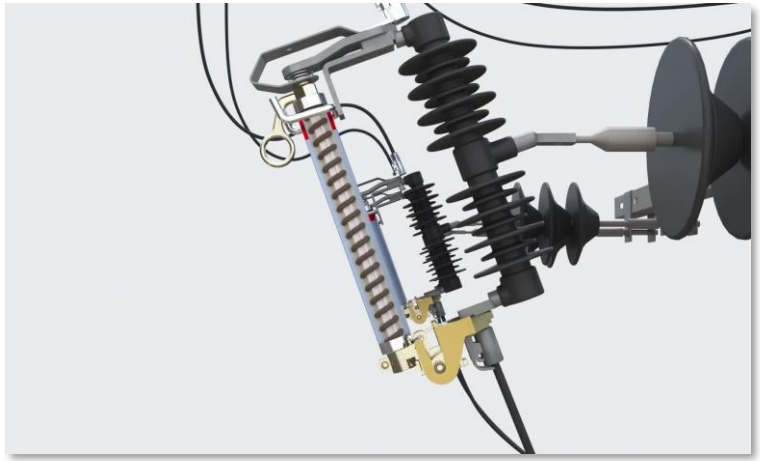
Replaced as one unit

No link assembly in the field



Adopted by KEPCO

Korea Electric Power Corporation

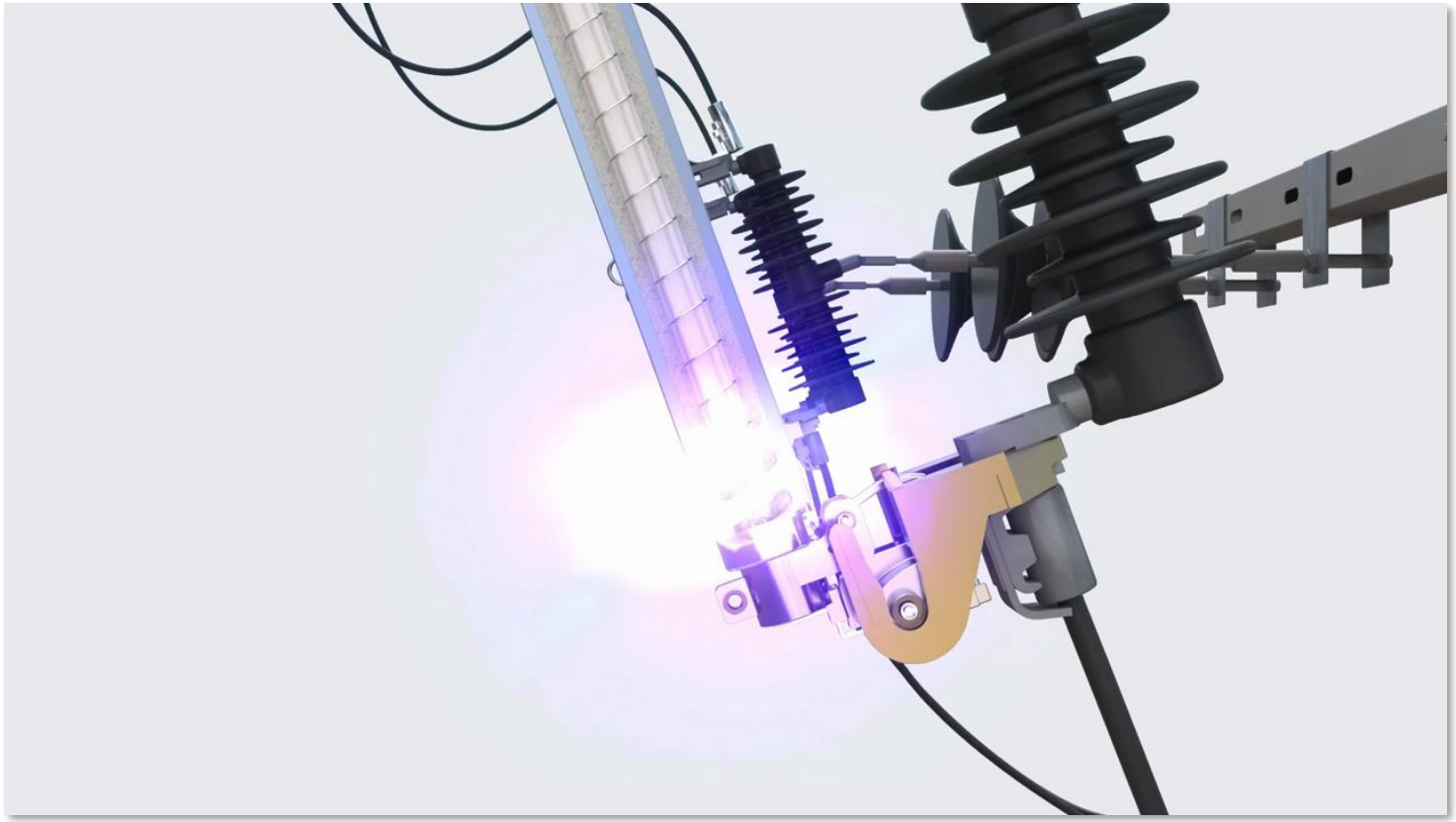


Exported overseas

New Zealand & Australia



The arc stays inside. The fuse drops out.



Animation · Reformtech

How the CLF fuse clears a fault

1

Fault occurs

Element melts and an arc forms

2

Arc absorbed

By silica sand inside the sealed tube

3

Cleared in ½ cycle

No arc or noise outside

4

Drops out

Striker releases the latch

Same conditions. Different outcome.

KEPCO side-by-side test · Gochang test lab · July 2015 · 13.2 kV

Expulsion fuse



Arc and noise. Fault cleared only when the upstream recloser operated.

Reformtech CLF fuse



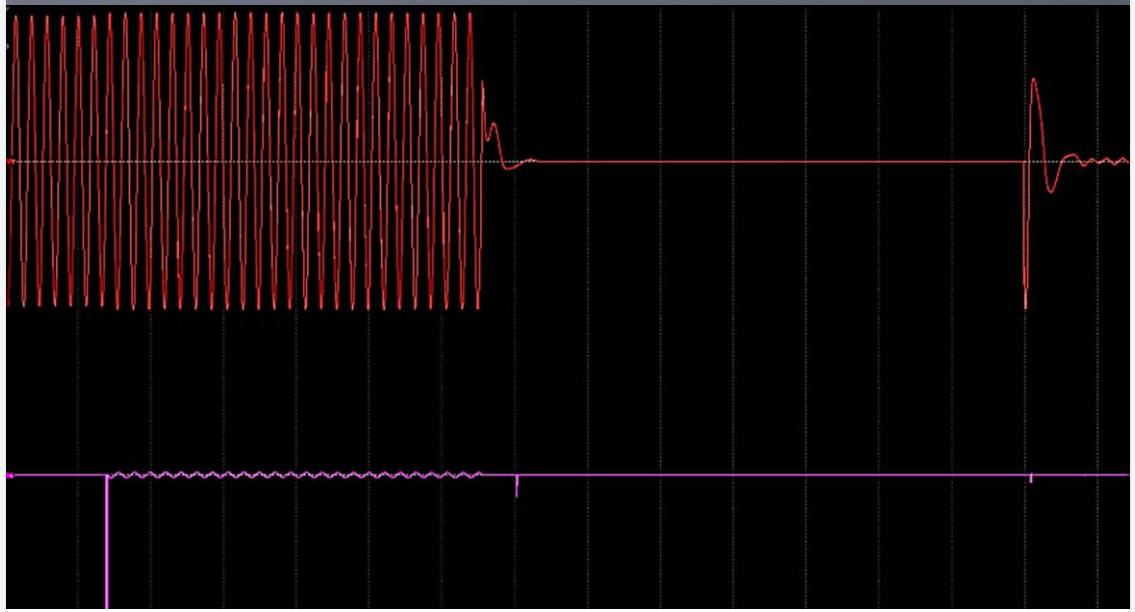
No arc, no noise. Fault cleared by the fuse itself.

200+ TESTS RUN BY KEPCO

KEPCO side-by-side test: recorded waveforms

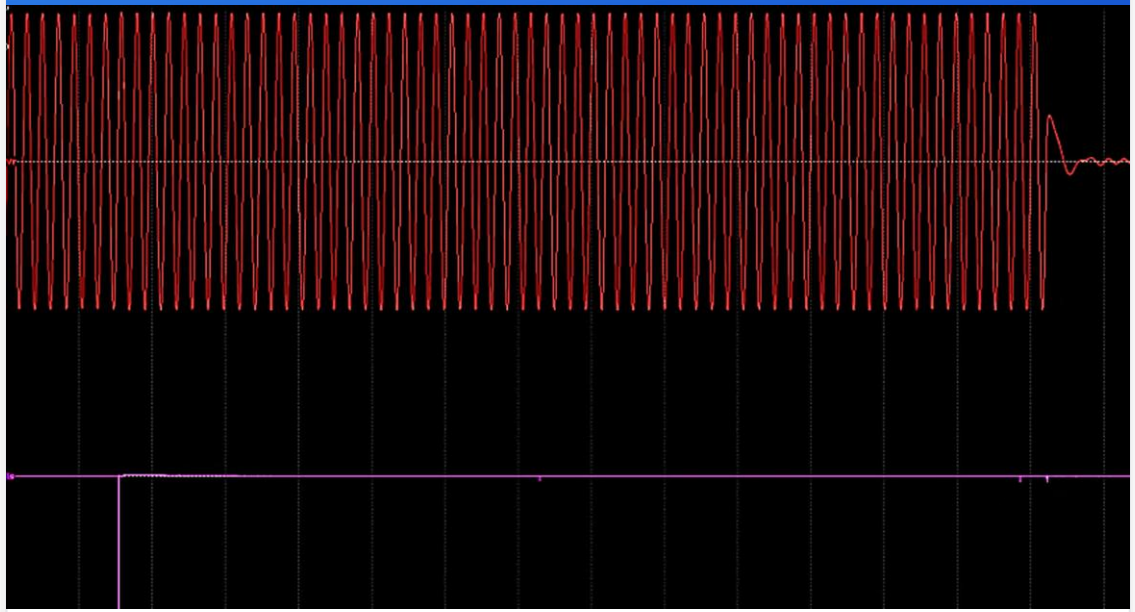
Backup for Q&A · Gochang test lab · July 2015 · 13.2 kV

Expulsion fuse



Fault current not interrupted by the fuse. Cleared only when the upstream recloser operated.

Reformtech CLF fuse



Fault current interrupted by the fuse. No recloser operation needed.

From pilot project to default choice

KEPCO adoption of the CLF fuse

MOU
Co-development MOU signed with KEPCO

2016
On-site technology development project completed

2017
Rated excellent; selected as KEPCO pilot product

2018
KEPCO notice of full-scale use

2023
Extended to mountain areas for fire safety

2024
Default choice in wildfire-prevention zones

2025
Purchase specs revised: 1A, 2A, 3A added

After the 2019 Goseong–Sokcho wildfire, KEPCO expanded use in high-risk areas.

GLOBAL TRACK RECORD

Proven in Korea, New Zealand & Australia

Tested

University of Canterbury High Voltage Lab, at Orion's request

Published

Results presented at the EEA Conference

Ordered

By Orion, South Island's largest power company

Also under evaluation in the United States, Chile, and Greece

"Fully confirmed and tested full-range Current Limiting COS Fuse"

University of Canterbury · EEA Conference paper



Korea



New Zealand



Australia





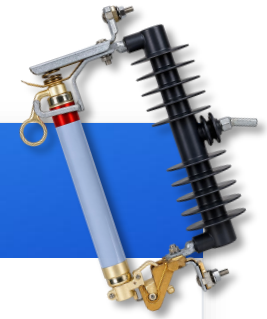
~30,000

units supplied to KEPCO

~20,000

units exported to New Zealand & Australia

Expulsion fuse vs. Reformtech CLF



	Expulsion fuse	Reformtech CLF
Arc & noise when operating	✗ Released outside	✓ **None, contained inside**
Fault energy (I^2t)	✗ Not limited	✓ **Significantly reduced**
Replacement	✗ Manual link assembly in the field	✓ **One complete unit**

IN SUMMARY

Why choose Reformtech CLF fuse?

Safer

No external arc or noise

More reliable

Faults cleared at the transformer, not by an upstream recloser

Proven

Adopted by KEPCO · independently validated in New Zealand

Let's partner in India!

We're looking for partners.

Visit us at

Booth [#12]